ACADEMIC CAPSTONE • ADVANCED WEB TECHNOLOGIES (AWT)

CampusConnect

Smart Campus Management System & Academic Workflow Portal

PHP 8.0+ (PDO) | MySQL / MariaDB (InnoDB) | Bootstrap 5.3.3 | Vanilla JavaScript (ES6+) | AJAX (Fetch API) | XML / DTD Validation

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CONTEXT & PROBLEM DEFINITION

The Challenge in Modern Campus Operations

Operational Breakdown

📁 Information Fragmentation

Lecture slides, circulars, and course schedules are scattered across third-party cloud drives, unvetted chat groups, and physical notice boards, resulting in missed student deadlines and communication gaps.

⏳ Manual Administrative Burden

Submitting assignments and managing deliverables over email causes lost attachments, lacks audit trails, and complicates teacher grading and structured student feedback delivery.

🚫 Zero Concurrency Controls

Campus event registrations and grievance handling lack centralized database constraints, resulting in double-bookings, capacity overruns, and unaddressed student issues.

> 🎤 **SPEAKER NOTE TO LEADER**
>
> *"In typical collegiate environments, academic resources are fragmented across messaging groups and emails. Faculty members manage submissions manually, which creates huge administrative friction. CampusConnect was engineered to solve this through a unified, database-driven portal."*

PROJECT OVERVIEW

CampusConnect: Two Distinct Role-Governed Portals

System Capabilities

👨‍🎓 Student Academic Portal

- **Personalized Dashboard:** Tracks pending coursework tasks and deadlines.
- **Digital Study Repository:** Asynchronous live search (AJAX Fetch API).
- **Coursework Submission:** Upload solutions (PDF/DOCX/ZIP) & view scores.
- **Campus Event Booking:** One-click symposium registration with duplicate prevention.
- **Grievance Portal:** Submit feedback with character counter & status tracker.

🛡️ Administrator Management Console

- **Institutional Analytics:** Real-time student counts, study files, and event stats.
- **User Management:** Audit accounts, filter roles, provision users safely.
- **Notice Board Publisher:** Publish, categorize, and pin circulars (CRUD).
- **Assignment & Grading Center:** Review student files, assign scores and feedback.
- **Event Management:** Set venue capacities, view and export attendee rosters.

🎤 SPEAKER NOTE TO LEADER

"CampusConnect is built on strict role-based access control. A student receives a productivity dashboard for their coursework, while an administrator receives an institutional management cockpit with full CRUD over all data."

ENGINEERING FOUNDATIONS

Technology Stack & Architectural Rationale

Decision Matrix

Layer	Technology Selected	Engineering Rationale & Benefit
Frontend UI	`Bootstrap 5.3.3`	Provides clean, responsive, accessible components with zero dependency bloat.
Client Scripting	`Vanilla JS (ES6+)`	Native Fetch API for instant AJAX search, input debouncing, and DOM manipulation.
Animation Demo	`jQuery 3.7.1`	Isolated interactive module demonstration satisfying academic syllabus requirements.
Backend Engine	`PHP 8.0+`	Strong typing, native session encryption, and clean modular include architecture.
Database Driver	`PHP Data Objects (PDO)`	Enforces 100% parameterized SQL prepared statements to eliminate SQL Injection.
Database	`MySQL / MariaDB (InnoDB)`	Relational integrity with foreign keys (`ON DELETE CASCADE`) and composite constraints.
Interchange	`XML / DTD Validation`	DOMDocument validates course schemas dynamically before rendering accordions.

🎤 SPEAKER NOTE TO LEADER

"We intentionally avoided bloated Node frameworks to deliver a clean, maintainable PHP 8 and MySQL stack. Using PDO ensures complete immunity against SQL injection, while Bootstrap 5 gives us sub-50ms rendering."

SYSTEM ARCHITECTURE

High-Level Flow & Execution Pipeline

Flowchart Pipeline

🌐

1. Client Tier

Browser dispatches HTTP navigation or Fetch AJAX queries.

⚙️

2. Apache & PHP 8

Gateway enforces CSRF tokens and session role checks (`require_role`).

🗄️

3. Dual-Data Tier

Prepared PDO queries to MySQL or DOM parsing on XML & DTD.

📊

4. Response Output

Renders sanitized HTML views or dispatches JSON payloads.

CORE DECISION LOGIC:

Query Execution Status evaluates on server → On **Success**, renders interactive dashboard or JSON data → On **Exception**, triggers safe contextual validation alerts (403/422) without exposing database stack traces.

🎤 **SPEAKER NOTE TO LEADER**

"When a user interacts with CampusConnect, Apache routes requests to the PHP 8 runtime. The backend verifies CSRF tokens and role permissions, queries MySQL via PDO or parses XML via DOMDocument, and dispatches pre-rendered HTML or JSON."

USER INTERFACE TIER

Frontend Engineering & Tri-State Theme Engine

UI/UX Subsystem

☀️ Light Mode

Off-white surfaces (`#f8fafc`), high-contrast charcoal typography (`#1e293b`), and restrained blue accents. Optimized for well-lit lecture halls.

🌙 Dark Mode

Sleek slate and charcoal backgrounds (`#0f172a` , `#1e293b`) minimizing eye strain during nighttime study sessions.

🔄 System Auto Mode

Dynamically synchronizes with the operating system theme using `window.matchMedia` with instant Cross-Tab synchronization.

⚡ Zero Flash of Unstyled Theme (FOUC) Solution

A lightweight inline script runs synchronously in the HTML `<head>` before page render, reading `localStorage` and injecting the theme attribute before the browser paints the DOM, completely preventing white flashes.

> 🎤 **SPEAKER NOTE TO LEADER**
>
> *"On the frontend, our theme switcher supports true 'System Auto' mode following the user's OS preference. We solved the annoying 'white flash' bug by executing a micro-script in the document head before the browser paints."*

SERVER TIER

Backend Architecture: Modular PHP 8 & Sessions

Core Engine

🧩 Modular Project Organization

- `config/`: Centralized PDO database manager with seamless SQLite testing fallback.
- `includes/`: Security guards (`require_role()`), CSRF tokens, and view components.
- `api/`: RESTful endpoints returning structured JSON.
- `actions/`: Self-contained transactional action handlers.

🔐 Cryptographic Session Lifecycle

- **Bcrypt Password Hashing:** Utilizes `password_hash()` with salt.
- **Session Fixation Mitigation:** Invokes `session_regenerate_id(true)` immediately upon successful authentication.
- **Cookie Hardening:** `HttpOnly` and `SameSite` flags prevent JavaScript access to cookies.

> 🎤 **SPEAKER NOTE TO LEADER**
>
> *"Our backend follows clean modular PHP conventions. Password authentication relies on Bcrypt hashing, and we immediately regenerate the session ID upon sign-in to neutralize session fixation attacks."*

PERSISTENCE TIER

Relational Database Architecture (MySQL InnoDB)

Normalized Schema

users
Credentials, role, department, roll number.

announcements
Circulars, categories, priority pinning.

resources
Study documents, file paths, size, MIME type.

assignments
Tasks, subjects, deadlines, max marks.

assignment_submissions
Uploads, notes, status, marks, feedback.

events
Symposiums, dates, venues, seat capacity.

event_registrations
Junction table linking students to booked events.

feedback
Grievances, resolution status, admin remarks.

RELATIONAL CONSTRAINTS: Composite unique constraint on `UNIQUE (event_id, student_id)` eliminates duplicate registrations. Cascading foreign keys preserve integrity across deletions.

🎤 SPEAKER NOTE TO LEADER

"Our database features 8 normalized tables with foreign key cascades and composite unique constraints. For instance, in event registrations, duplicate bookings are rejected at the database engine level."

INTERACTIVE CLIENT SCRIPTING

Asynchronous AJAX Search Engine (Fetch API + JSON)

Real-Time Search

🔍 The 3-Step Search Pipeline

- **1. Keystroke Debounce:** Client listens on `input` with a 300ms timer to prevent server flooding.
- **2. Fetch API Dispatch:** Asynchronously requests `api/search.php?q=Web`.
- **3. Dynamic DOM Ingestion:** Injects formatted cards into container in-place with zero page reload.

📦 Structured JSON Payload

```
{
  "status": "success",
  "count": 1,
  "data": [{
    "id": 1,
    "title": "AWT Lecture Notes & PDO Guide",
    "subject": "Web Technology",
    "formatted_size": "2.4 MB",
    "time_ago": "5 days ago"
  }]
}
```

🎤 SPEAKER NOTE TO LEADER

"Rather than reloading the whole page when searching for lecture notes, our JavaScript debounces the keystrokes and queries our JSON endpoint via the Fetch API, updating the results instantly."

DATA INTEROPERABILITY

XML & DTD Curriculum Validation Subsystem

Schema Validation

📄 Document Type Definition (courses.dtd)

- Defines rigid structural rules for course cataloging.
- Enforces nested elements: `title`, `credits`, `instructor`, `modules`, and `prerequisites`.
- Validates attribute requirements like academic year and credit limits.

⚙️ PHP DOM Parsing Engine (courses.php)

- **Validation:** Executes `$dom->validate()` ensuring zero schema violations.
- **Transformation:** Uses `SimpleXMLElement` to extract modules and topics.
- **Display:** Generates interactive Bootstrap accordions displaying module breakdown and teaching hours.

🎤 SPEAKER NOTE TO LEADER

"To demonstrate XML interoperability in our curriculum, we engineered an XML course catalogue validated against a strict DTD schema. PHP's DOMDocument validates the document rules before displaying it."

APPLICATION HARDENING

Enterprise Security Architecture (OWASP Protections)

Defense-in-Depth

🗡️ SQLi Elimination

100% Parameterized PDO Statements. Zero dynamic SQL concatenation ensures user input is strictly treated as query parameters.

🛡️ XSS Defense

Contextual HTML Escaping. All user-supplied output is passed through our `e()` helper utilizing `htmlspecialchars()`.

🎫 CSRF Tokens

Cryptographic Session Tokens. Every state-changing form includes a secure token verified via `validate_csrf()`.

🔑 Bcrypt Encryption

Passwords stored with cryptographic salt utilizing PHP's `password_hash()`.

📁 Path Traversal Guards

`download.php` verifies canonical paths via `realpath()` inside the uploads folder.

🔒 Role Enforcement

Server-side access checks (`require_role`) strictly prevent privilege escalation.

> 🎤 **SPEAKER NOTE TO LEADER**
>
> *"Security was engineered into the foundation: prepared statements against SQL injection, strict HTML escaping against XSS, cryptographic CSRF tokens, and path traversal guards on file downloads."*

DEMONSTRATION FLOW

Step-by-Step Live Demonstration Walkthrough

Live Evaluation

1 Student Workflow (aarav.sharma@campusconnect.edu)

- **Login & Personalized Greeting:** Shows roll number, pending tasks, and recent notices.
- **Live AJAX Search:** Search for "Web" lecture slides with zero page refresh.
- **Coursework Submission:** Submit written notes and attach a solution archive.
- **Event Booking:** Book a seat for HackCampus 2026; seat counter updates live.
- **Feedback Submission:** Watch the character counter while submitting an inquiry.

2 Administrator Workflow (admin@campusconnect.edu)

- **Institutional Analytics:** Review student statistics, materials, and pending alerts.
- **Submissions Evaluation:** Inspect Aarav's uploaded solution; enter grade and feedback.
- **Attendee Roster:** View and export the registered participant list for HackCampus.
- **Notice Board Publisher:** Publish a new circular and pin it to the top.
- **Grievance Resolution:** Reply to student feedback; updates status to 'Reviewed'.

🎤 SPEAKER NOTE TO LEADER

"Now I will switch to our live environment. First, I will demonstrate the student journey: live search, assignment upload, and event registration. Then, I will log into the admin console to evaluate submissions and resolve queries."

TESTING & VERIFICATION

Quality Assurance & Automated Verification Results

QA Verification

🧪 Automated QA Test Suite (tests/test_suite.php)

- PASS Database PDO Connection
- PASS All 8 Required Tables & Constraints Exist
- PASS Admin & Student Bcrypt Password Verification
- PASS Duplicate Event Registration Constraint Enforcement
- PASS CSRF Token Cryptographic Validation
- PASS XSS Sanitization Helper e()
- PASS XML Course Catalogue & DTD Schema Validation
- PASS Study Resource Query Search Execution

📊 Static Linting & Browser Results

- **Zero Syntax Errors:** All 34 PHP files passed `php -l` linting with 0 warnings.
- **DTD Schema Conformance:** `courses.xml` confirmed valid against DTD schema.
- **Browser Automation:** Light/Dark theme switching, responsive layouts, and form validations verified end-to-end.

🎤 SPEAKER NOTE TO LEADER

"Before concluding development, we ran our custom automated test suite. Every subsystem—from database constraints and password verification to XML schemas and CSRF defenses—passed with 100% success."

CURRICULUM ALIGNMENT

Web Development Syllabus Coverage Matrix

Academic Mapping

🎨 HTML5, CSS3 & Bootstrap

Semantic landmarks (`<header>`, `<main>`, `<footer>`), responsive grid, CSS variables, Light/Dark/System themes, and an educational marquee sandbox.

⚡ JavaScript, AJAX & jQuery

Native Fetch API, 300ms input debouncing, real-time client validation, character counters, and jQuery `.slideToggle()` interactive animation.

🦣 PHP 8, MySQL & XML

Role-based sessions, Bcrypt hashing, PDO prepared statements, InnoDB foreign keys with cascading deletions, and XML DTD validation.

🎤 **SPEAKER NOTE TO LEADER**

"CampusConnect fulfills all objectives of our Web Development syllabus. Every curriculum module—from semantic HTML5 landmarks and CSS tokens to AJAX Fetch requests, XML DTD validation, and PHP PDO transactions—is implemented in real working features."

SUMMARY & FUTURE SCOPE

Conclusion, Business Value & Future Roadmap

Next Milestones

🏆 What CampusConnect Delivers Today

- **Centralized Hub:** Eliminates information fragmentation across the campus.
- **Zero Artificial Gimmicks:** Real database records, production-grade security, and accessible UI.
- **Portable Deployment:** Instantly runnable on any standard XAMPP environment.

🚀 Future Enhancement Roadmap

- **Email Reminders:** Automated SMTP notifications for pending assignment deadlines.
- **Automated PDF Badges:** Dynamic certificate generation for symposium attendees.
- **Real-Time Messaging:** WebSocket-powered student-to-instructor advisory channels.

🎤 SPEAKER NOTE TO LEADER

"In summary, CampusConnect provides a robust, scalable, and secure platform that solves real campus operational challenges. Thank you, and I am now ready to take your questions and demonstrate the code!"